

Definite Integrals and the Fundamental Theorem — Diagnosing Bounds, Sign and Variable-of-Integration Errors

Three slips account for most lost points on FTC free-response work. Every problem below contains one of them.

- **Bound order and sign:** $\int_a^b f = -\int_b^a f$. Bounds that run backwards, a variable sitting at the *lower* bound, and a substitution that changes the variable but not the bounds all show up as a sign or a scale that is simply wrong.
- **Evaluate, then subtract as a whole:** $\int_a^b f = F(b) - F(a)$. The parentheses around $F(a)$ are not decoration — the minus reaches every term inside them.
- **The variable of integration is a dummy:** it is consumed by the integral. It cannot survive into the answer of a definite integral, and it is not something you substitute a bound into.
- In each problem: name the error in words first, then show corrected work.

1. Find and fix. Evaluate $\int_5^2 2x \, dx$.

Jae writes “the antiderivative is x^2 , so the answer is $5^2 - 2^2 = 21$.”

Say in one sentence what Jae did to the bounds, then give the correct value.

Answer: _____

2. Find and fix. Evaluate $\int_1^3 (3x^2 - 4) dx$.

Ravi finds the antiderivative correctly, then writes $27 - 12 - 1 - 4 = 10$.

- (a) Write the antiderivative $F(x)$ Ravi should be using.
- (b) Name the punctuation Ravi lost, then evaluate the integral correctly.

Answer: _____

3. Find and fix. Evaluate $\int_0^2 (3t^2 + 2) dt$.

Noor writes “ t runs up to 2, so put 2 in for t ”: $3(2)^2 + 2 = 14$.

Explain what Noor skipped between the integrand and the bounds, then evaluate the integral correctly.

Answer: _____

4. Find and fix. A continuous function f satisfies $\int_1^6 f(x) dx = 10$ and $\int_1^4 f(x) dx = 3$.

Asked for $\int_6^4 f(x) dx$, a student subtracts and answers 7.

(a) Say which integral the value 7 actually is.

(b) Give the correct value of $\int_6^4 f(x) dx$.

Answer: _____

5. Find and fix. Evaluate $\int_0^{\pi/3} \sin x dx$.

Lena writes “the antiderivative of $\sin x$ is $\cos x$ ” and reports $\cos(\pi/3) - \cos(0)$.

State the correct antiderivative, say what Lena’s version does to the answer, and evaluate the integral correctly.

Answer: _____

6. Find and fix. Let $G(x) = \int_x^5 (t^2 + 1) dt$.

Kwame writes $G'(x) = x^2 + 1$ and reports $G'(2) = 5$.

- (a) Rewrite $G(x)$ with the variable at the upper bound, and say what that does to the sign.
(b) Compute $G'(2)$ correctly.

Answer: _____

7. Find and fix. Evaluate $\int_0^2 x(x^2 + 1)^3 dx$ by substitution.

Ines sets $u = x^2 + 1$, gets $\frac{u^4}{8}$, and then evaluates it at the original bounds, reporting 2.

- (a) Convert the bounds to u and state the new interval.
(b) Evaluate the integral correctly.

Answer: _____

8. Explain.

- (a) Using $\int_a^b f = F(b) - F(a)$, explain why swapping the bounds negates a definite integral. Say what that means for the signed area.
- (b) Explain why the letter used for the variable of integration can never appear in the value of a definite integral, and why it *can* appear in the derivative of an accumulation function.
- (c) Describe what goes wrong when a substitution changes the variable but not the bounds, and give a habit that prevents it.